



“I’ll Be Back”: A Taxonomy of Death and Rebirth in Platformer Video Games

Marjorie Ann M. Cuervo

DePaul University
Chicago, IL 60614, USA
mcuerdo@mail.depaul.edu

Edward F. Melcer

University of California, Santa Cruz
Santa Cruz, CA 95064, USA
eddie.melcer@ucsc.edu

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI’20 Extended Abstracts, April 25–30, 2020, Honolulu, HI, USA.
© 2020 Copyright is held by the owner/author(s).
ACM ISBN 978-1-4503-6819-3/20/04.
DOI: <https://doi.org/10.1145/3334480.3382863>

Abstract

Failure, often represented through death, is a central aspect of almost every video game. In-game death can drive player perceptions of difficulty and greatly impact the core player experience; however, there is surprisingly limited amounts of research examining how games actually handle this occurrence. We posit that this is a rich, underexplored space with significant implications for player experience and the design of many games. This paper presents our initial exploration into the space of player death and rebirth through the creation of a generalized taxonomy developed from 62 recent platformer games. Our taxonomy consists of five key dimensions: 1) *obstacles*, 2) *death conditions*, 3) *aesthetics*, 4) *changes to player progress*, and 5) *respawn locations*.

Author Keywords

Taxonomy; Platformers; Games; In-Game Death; Grounded Theory; Player Experience.

CSS Concepts

- **Human-centered computing~Human computer interaction (HCI)**

Introduction

Failure in video games, often represented through death, is a central element of the player experience [26]. For instance, failure is well documented to have substantial positive and negative impacts on game enjoyment [27]. It is also intrinsically linked to

challenge, and consequently a major factor in the experience of game flow [35]. While there have been various approaches to examining and manipulating this critical aspect of the gaming experience—e.g., dynamic difficulty adjustment [15, 23], challenge design [5, 59, 70] and modeling [66]—research understanding failure explicitly related to handling in-game death is surprisingly limited [47]. We propose that how video games deal with in-game death is a rich, underexplored space with significant implications for player experience and related techniques such as dynamic difficulty adjustment. With that in mind, we created a taxonomy of player death and rebirth in platformer games to better understand the current design space.

Platformers are a video game genre where players typically control a game character to jump between platforms while avoiding obstacles. We focused on platformers as they are generally designed around constant player death and respawning in pursuit of a goal. They are also notorious for being difficult – with many such games leading to the creation of the term "Nintendo hard", which describes the extreme difficulty of games from the Nintendo Entertainment System era [18]. The fairness of difficulty in Nintendo hard games has been called into question in recent years [39], highlighting that such designs can sometimes serve more to infuriate players through frequent unfair death rather than provide an appropriate challenge. A taxonomy of death and rebirth in this space provides a tool to compare death and respawning mechanics across games—helping to elucidate common design choices in existing platformer games, as well as aid exploration of which mechanics and designs may impact on player experience.

In this paper, we set out to understand the essential features and mechanics around death in video games. We employed grounded theory to analyze existing platformer games. Since there is a relatively limited amount of literature that address the actual mechanics around in-game death, games are a more useful

information source [2] and served as our main source of data for analysis. We also provide a ludography for the 62 games in our dataset; therefore the reference for a game will be prefixed with a "G" (e.g., [G7]).

Background

Game Taxonomies

Taxonomies provide a means to organize and classify concepts [3]. In terms of games, there have been a substantial number of taxonomies and frameworks ranging from general classifications of games themselves [1, 17, 69] to various aspects of games—such as core mechanics [61], bugs [40], embodiment [45], player modeling [64], and external factors [43]—to specific genres of games, e.g., serious games [42, 54], games for dementia [10, 44], exertion games [49], affective games [38, 19], idle games [2], and games and simulations [33] to name a few. Of particular relevance to this work is the framework created by Smith and Whitehead [65] for analyzing 2D platformer levels. Their framework consists of *components* in the form of platforms, obstacles, movement aids, collectible items, and triggers; as well as *structural representation* for how the components fit together. We utilize many of these concepts in our coding scheme and taxonomy; however their framework ultimately focuses on rhythm and pacing to evoke challenge, rather than death.

In-Game Death

Death is also an intrinsic part of gameplay [50] and the player experience [34]. Players can be impacted by death in games when they are attached to characters at some emotional level or self-identify with the goals or events in the game [4, 48, 55, 60]. For instance, when players are immersed in gameplay, the risk of death produces anxiety and encourages more careful decision-making, but can also evoke strong emotions from players—resulting in enjoyment and positive player experience despite the frustration that comes with the territory [4].

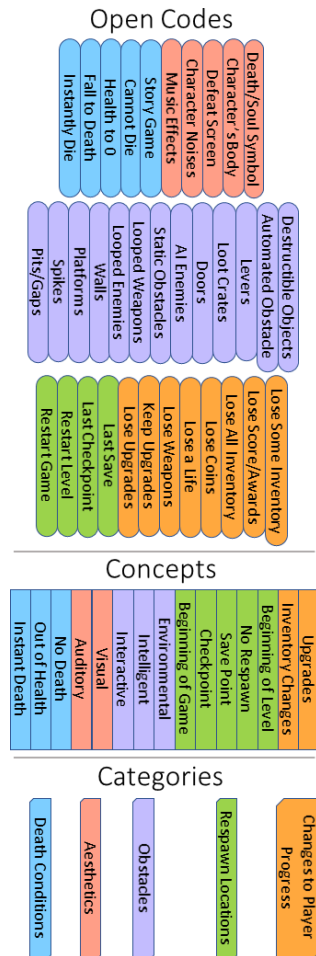


Figure 1: The coding process with 1) open coding, 2) related into concepts using axial coding, and 3) grouped using selective coding to create taxonomy categories.

With respect to platformer games, their notoriously difficult level design can often result in frequent occurrences of player death [18]. These instances of in-game death generally impede player progress, e.g., through the loss of inventory items, achievements, or game functionality [4]. At even greater extremes, in-game death can completely reset player progress such as through the popular high-risk death mechanic, permadeath—i.e., the permanent in-game death of a playable character (c.f., [13])—which forces the player to restart the entire game upon dying. Notably, these repetitive deaths have been found to increasingly reduce player enjoyment, as each death occurrence compounds as an evaluation of the insufficiency of a player's skills [21]. As such, it is critical to examine the features that comprise in-game death for a broader understanding of its effects on player experience.

Grounded Theory

Grounded Theory methodology (GTM) is commonly used to explore new domains [2, 20]. For instance, within games, GTM has been used to analyze idle games [2], game immersion [6], and cooperative communication mechanics [67]. The GTM process starts with data collection, gradually building up categories and forming a theory, before linking that theory to previous literature at the end [22]. This effectively enables a researcher to simultaneously analyze a body of artifacts (in this case platformer video games) and develop a theory about what elements of these artifacts are salient [36]. For the creation of our taxonomy, we adopted the Constructivist Grounded Theory flavor of GTM [9]. This GTM variant frames the researcher as co-creating meaning within the domain they are studying [8], focusing on providing lenses for analysis rather than a single objectively correct domain model [57].

Methodology

We conducted an analysis of 62 platformer games in order to identify the essential features and differing

mechanics around death and respawning in platformer games. We utilized a constructivist grounded theory approach [9] that started with an iterative process of selecting platformer games. Games were then analyzed and coded by watching videos of gameplay, and playing when necessary. Specifically, we employed open coding and conceptual memoing to identify the initial death and respawning concepts. Axial coding—i.e., identifying relationships among the open codes and initial concepts [2]—was then employed to determine our initial set of categories. Finally, selective coding—i.e., integrating initial categories to form a core category that describes the data comprehensively [2]—was used to determine the final categories of our taxonomy (see Figure 1).

Search Strategy

In order to obtain an accurate corpus of recent popular platformer games, we utilized the digital games distribution platform, *Steam*, for our search of games explicitly tagged as "Platformer". In an effort to observe characteristics of recent platformers, we restricted the search to video games that were released within the period of January 2018 to May 2019. We then selected the top twenty or so games with the highest number of user reviews for positive, mixed, and negative ratings. A total of 62 games were selected: (1) *positively-reviewed*, 21 collected; (2) *mixed-reviewed*, 22 collected; and (3) *negatively-reviewed*, 19 collected.

Analysis Procedure

Phase 1: Observations of Death & Rebirth Mechanics

We examined each game individually by watching a playthrough online or, if not available, obtaining the game to play it ourselves. We recorded each game's approach to handling player death with information on what conditions result in death, where players were respawned, what was lost and gained, obstacle types, and visual and auditory representations of death.

Phase 2: Open, Axial, and Selective Coding

We started this phase by performing open coding on our observations of the 62 platformer games from

DEATH & REBIRTH



Figure 2: The Taxonomy of Death & Rebirth in Platformer Games. *Obstacles* are cause *Death Conditions* to be met. Death is conveyed through *Aesthetics* and causes *Changes to Player Progress*. Players are then reborn into *Respawn Locations* where they again attempt to overcome obstacles as the cycle repeats.

phase 1. Axial coding was then employed to identify a set of emerging concepts and initial categories around death and respawning. This was followed by multiple iterative discussion sessions to explore the relationships between the open codes, emergent concepts, and initial categories—resulting in selective coding of the 5 key categories for our taxonomy (see Figure 1).

A Taxonomy of Death & Rebirth

Based on the concepts and mechanics that emerged from our analysis, we formed the **Taxonomy of Death & Rebirth in Platformer Games** (see Figure 2). Our taxonomy describes 5 major aspects of the cyclical process of death and rebirth: (1) *Obstacles*, which are the cause of (2) *Death Conditions* being met and resulting in player death depicted through (3) *Aesthetics* as well as causing (4) *Changes to Player Progress* before being reborn at (5) *Respawn Locations* to repeat the entire process. While not every game follows this process exactly (e.g., characters cannot die in some games [G23, G26-27, G43-44]), this taxonomy provides the high-level structure necessary to understand, break down, and categorize the process of death and rebirth among a variety of platformer games.

Obstacles

Obstacles in platformers present challenges and difficulties for players to overcome [65, 66, 70]. They are also critical in existing literature for analyzing [65], dynamically adjusting [23], and generating [14, 62] platformer levels. The effects that obstacles have on player progress can either disrupt or encourage flow in gameplay [24, 41]. Ultimately, they are key factors that lead to player death [70]. To analyze the role of obstacles in platformers, we determined what type of obstacle was the most prominent in each observed game. We identified three notably different types of obstacles that could lead to player death:

- *Intelligent* obstacles are objects in the game that actively attempt to kill the player, and their

movements/actions respond in real-time to player actions. Examples include enemy characters that follow a player and deadly moving objects, such as projectiles that aim towards the player's location.

- *Environmental* obstacles are directly part of the game environment and can lead to player death. They can either be static obstacles that don't move or automated obstacles that move in fixed ways. Examples of static environmental obstacles include immovable components of the level, such as spikes, pits, platforms, and walls. Examples of automated environmental obstacles include moving platforms, rising lava, or enemy characters that follow fixed paths, looping their movements.
- *Interactive* obstacles are objects in the game that can be activated or interacted with by the player. Examples of these include doors, levers, destructible objects, and treasure chests.

Death Conditions

Platformer games heavily feature player death, which has been shown to evoke both positive [4] and negative [21] player experiences. As such, the death conditions and mechanics prevalent in these games are critical elements of the taxonomy. Specifically, we identified three distinct types of death conditions:

- *Instant Death* describes games where the character dies immediately from a single injury.
- *Out of Health* describes games where the life of a character is tied to a health bar. When the player runs out of health, the character dies. While health is always finite in this paradigm, players are given far more chances to escape death after a mistake.
- *No Death* describes games where death is not possible during gameplay, instead focusing on level design to create a loss of progress (e.g., [G26]).

Aesthetics

Although there are a number of definitions and interpretations of game aesthetics, aesthetics in relation to this taxonomy refers purely to the sensory phenomena that players encounter in the game [52]. Aesthetics can greatly impact player emotion and the overall gaming experience [29, 30, 51, 58], and therefore is an important category to consider for the design of in-game death. Specifically, we focus on the different variations of *Visual* and *Auditory* aesthetics that occur during in-game death.

Visuals are a critical aspect of the aesthetic experience, where even fundamental elements, such as shape and color, can have a substantial impact on player emotions and overall experience [29, 46, 52, 53, 68]. With respect to visual aesthetics around in-game death, we observed that visual changes were primarily focused on character appearance and/or use of death screens. Iconography, such as blood, skulls, and souls, was also often used to indicate character death, and could remain in the environment through multiple death and rebirth iterations to indicate where the player had previously died.

Audio, in the form of music and sound effects, is another critical aspect of the aesthetic experience. For instance, both music and sound effects have been shown to impact player immersion and emotional response [28, 30, 51, 58]. In the corpus, a number of different sound effects were employed during and immediately after death, such as cries or grunts, squishing noises, and electronic sounds. Music was also often modified, such as by abruptly stopping the background music or playing unique melodies.

Changes to Player Progress

After a player's character dies, aspects of their progress are either retained or lost. Changes to the player's progress are important to consider, as these types of changes have been shown to impact various aspects of player experience [37] and lead to strong emotional

responses [4]. While these changes do take on a variety of forms, in an abstract sense, they can be categorized as *Upgrades* and *Inventory Changes*.

UPGRADES

Many platformers enable retention of earned upgrades for a character after death. Examples of preserved upgrades observed in our corpus include power-ups, weapons, skill levels, and achievements. Prior work has shown that upgrades can have significant impact on enjoyment [32] and challenge [37], allowing players to customize their overall gaming experience [16].

INVENTORY CHANGES

Inventory systems in platformer games are another important feature as they contain various explicit indicators of player progress, such as currency, lives, and items. We observed that the changes to inventory ranged from players keeping all of their inventory after death to losing some or all of it. Notably, how much inventory is maintained or lost alters the consequence of in-game death for the player, and ultimately impacts the overall gameplay [31] and game experience [7].

Respawn Locations

When there is death, there is also respawning in platformer games, i.e., when the character is brought back to life to continue gameplay. However, where the player can reappear varies wildly. Poor use of respawn locations (e.g., too far away or too directly into action) can lead to negative player experience [11], and therefore is another important taxonomy category. We observed five distinct types of respawn locations:

- *Beginning of Game* is when players are directly sent to the game's initial menu screen upon death, and have to restart the entire game. Consequently, players often replay all or some sections of the game. This respawn location usually means that it requires more of a player's time to attempt completing the game. One popular subgenre of

games that focus entirely around this mechanic is permadeath [13].

- *Beginning of Level* occurs in games that are explicitly split into distinct levels or stages. When players die, they are respawned at the beginning of the level that they failed to successfully complete. Usually, this respawn location also means that any character progress achieved in the failed level is similarly lost upon death.
- *Checkpoints* occur when the character reaches a specific location in the game that *automatically* saves their progress. This was the most popular respawn mechanic in the corpus (30 out of 62 games), and most platformers with checkpoints increased their difficulty by expanding the distance between checkpoints as the game progressed. I.e., checkpoints break down levels into segments with smaller challenges for the player to overcome.
- *Save Points* differ from checkpoints in that they are consciously activated by the player. When players die, they are respawned at those exact locations where the save point was activated. As a result, players are given a greater level of autonomy and control in games that have save points.
- *No Respawn* mechanics were present in the same games that featured *No Death* in their gameplay. The absence of death means there is no requirement for the player to respawn into the game.

Discussion

We observed various platformer games to develop death and rebirth concepts for our Taxonomy of Death & Rebirth in Platformer Games. Our taxonomy highlights that there are a substantial number of mechanics, design decisions, and aesthetics that go into death and respawning in games, despite the limited amount of literature examining these features.

Guiding Dynamic Difficulty Adjustment Techniques

Game designers must focus on the balance between inspiring confidence in players and providing sufficient challenge [27]. Techniques such as Dynamic Difficulty Adjustment (DDA) attempt to do this by continuously adapting a game's difficulty to their current skill level [25, 63, 70]—keeping players in a constant state of flow [15]. There is evidence that dynamically adjusting components in games affects their perceived difficulty [15, 25, 70] and can decrease the rate of rage-quitting by increasing their perceived self-efficacy [12, 17]. However, current DDA techniques have been criticized for an overemphasis on designer intuition rather than actual play patterns [25]. While related methods have utilized machine learning [25], our taxonomy presents new possibilities since in-game death is inherently linked to challenge. Our taxonomy presents a tool grounded in the design of commercial platformer games to categorize different design possibilities for death and rebirth. It then provides new elements of platformer design for dynamic adjustment. For instance, dynamically adjusting how far backwards players respawn, how much progress is maintained, and the overall conditions for death are all approaches that have not been explored deeply in current DDA literature, but are highlighted in our taxonomy.

Conclusion and Future Work

We utilized grounded theory to develop a Taxonomy of Death & Rebirth in Platformer Games. Our goal is to provide a means for game designers and researchers to better design/analyze how games handle death. We identified 5 key categories as the basis of our taxonomy: *Obstacles*, *Death Conditions*, *Aesthetics*, *Changes to Player Progress*, and *Respawn Locations*. Further studies should be conducted to more deeply understand how concepts in our taxonomy impact crucial aspects of player experience such as game flow, engagement, challenge, autonomy, and self-efficacy. For instance, some death and rebirth mechanics could improve the perceived level of control to help satisfy a player's need for autonomy [56].

References

- [1] Espen Aarseth, Solveig Smedstad, and Lise Sunnana. 2003. A multidimensional typology of games. In *Proceedings of the 2003 DiGRA International Conference: Level Up*. DiGRA, Utrecht, The Netherlands. <http://www.digra.org/wp-content/uploads/digital-library/05163.52481.pdf>
- [2] Sultan Alharthi, Olaa Alsaedi, Zachary Toups, Theresa Tanenbaum, and Jessica Hammer. 2018. Playing to Wait: A Taxonomy of Idle Games. 1–15. DOI: <http://dx.doi.org/10.1145/3173574.3174195>
- [3] Alissa Antle and Alyssa Wise. 2013. Getting down to details: Using learning theory to inform tangibles research and design for children. *Interacting with Computers* 25 (01 2013), 1–20. DOI: <http://dx.doi.org/10.1093/iwc/iws007>
- [4] Julia A. Bopp, Elisa D. Mekler, and Klaus Opwis. 2016. Negative Emotion, Positive Experience?: Emotionally Moving Moments in Digital Games. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '16)*. ACM, New York, NY, USA, 2996–3006. DOI: <http://dx.doi.org/10.1145/2858036.2858227>
- [5] Michael Brandse. 2017. The Shape of Challenge: Using Affordance Design to Create Challenge Within. In *Design, User Experience, and Usability (HCI International 2017)*. Springer, Gewerbestrasse 11, 6330 Cham, Switzerland, 362–276. DOI: http://dx.doi.org/10.1007/978-3-319-58637-3_29
- [6] Emily Brown and Paul Cairns. 2004. A grounded investigation of game immersion. In *CHI'04 extended abstracts on Human factors in computing systems* (pp. 1297–1300). ACM.
- [7] Marcus Carter, Martin Gibbs, and Greg Wadley. 2013. Death and dying in DayZ. In *Proceedings of The Australasian Conference on Interactive Entertainment: Matters of Life and Death (IE 2013)*. ACM, New York, NY, USA. DOI: <http://dx.doi.org/10.1145/2513002.2513013>
- [8] Kathy Charmaz. 2000. *Handbook of Qualitative Research* (2nd ed.). Sage, Chapter Grounded Theory Methodology: Objectivist and Constructivist Qualitative Methods, 509–535.
- [9] Kathy Charmaz. 2006. Constructing Grounded Theory: A Practical Guide through Qualitative Analysis. SAGE, London, Thousand Oaks, New Delhi.
- [10] Dormann Claire. 2016. Toward Ludic Gerontechnology: a Review of Games for Dementia Care.. In *1st International Joint Conference of DiGRA and FDG (DiGRA/FDG '16)*. DOI: <http://dx.doi.org/10.1145/3311350.3347173>
- [11] Delwin Clarke and P. Robert Duimering. 2006. How computer gamers experience the game situation: a behavioral study. *Computers in Entertainment (CIE) - Theoretical and Practical Computer Applications in Entertainment* 4, 3 (2006). DOI: <http://dx.doi.org/10.1145/1146816.1146827>
- [12] Thomas Constant and Guillaume Levieux. 2019. Dynamic Difficulty Adjustment Impact on Players' Confidence. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '19)*. ACM, New York, NY, USA. DOI: <http://dx.doi.org/10.1145/3290605.3300693>
- [13] Amra Copcic, Sophie McKenzie, and Michael Hobbs. 2013. Permadeath: A review of literature. In *IEEE International Games Innovation Conference (IGIC 2013)*. IEEE. DOI: <http://dx.doi.org/10.1109/igic.2013.6659156>
- [14] Steve Dahlskog and Julian Togelius. 2012. Patterns and procedural content generation: revisiting Mario in world 1 level 1.. In *Workshop on Design Patterns in Games (DPG '12)*. ACM, New York, NY, USA. DOI: <http://dx.doi.org/10.1145/2427116.2427117>

- [15] Alena Denisova and Paul Cairns. 2015. Adaptation in Digital Games: The Effect of Challenge Adjustment on Player Performance and Experience. In *Annual Symposium on Computer-Human Interaction in Play (CHI PLAY '15)*. ACM, New York, NY, USA, 97–101. DOI:<http://dx.doi.org/10.1145/2793107.2793141>
- [16] Alena Denisova and Elliott Cook. 2019. Power-Ups in Digital Games: The Rewarding Effect of Phantom Game Elements on Player Experience. In *Annual Symposium on Computer-Human Interaction in Play (CHI PLAY '19)*. DOI:<http://dx.doi.org/10.1145/3311350.3347173>
- [17] Christian Elverdam and Espen Aarseth. 2007. Game Classification and Game Design: Construction Through Critical Analysis. *Games and Culture* 2, 1 (2007). DOI: <http://dx.doi.org/10.1177/1555412006286892>
- [18] M Enger. 2012. What is "Nintendo hard"? (2012). <http://bnbgaming.com/2011/02/08/what-is-nintendo-hard/>.
- [19] Kiel M. Gilleade and Alan Dix. 2004. Using frustration in the design of adaptive videogames. In *2004 ACM SIGCHI International Conference on Advances in Computer Entertainment Technology (ACE '04)*. ACM, New York, NY, USA, 228–232. DOI: <http://dx.doi.org/10.1145/1067343.1067372>
- [20] Barney G. Glaser. 2017. Discovery of grounded theory: Strategies for qualitative research. Routledge.
- [21] W.M. Hoogen, Karolien Poels, Wijnand Ijsselstein, and Yvonne De Kort. 2012. Between Challenge and Defeat: Repeated Player-Death and Game Enjoyment. *Media Psychology* 15 (10 2012), 443–459. DOI: <http://dx.doi.org/10.1080/15213269.2012.723117>
- [22] Nathan Hook. 2015. *Game Research Methods*. ETC Press, Pittsburgh, PA, USA, Chapter Grounded Theory, 309–320.
- [23] Robin Hunnicke. 2005. The case for dynamic difficulty adjustment in games. In *In Proceedings of the International Conference on Advances in Computer Entertainment Technology (ACE '05)*, Vol. 265. 429–433. DOI:<http://dx.doi.org/10.1145/1178477.1178573>
- [24] Aaron Isaksen, Dan Gopstein, Julian Togelius, and Andy Nealen. 2015. Exploring Game Space Using Survival Analysis. In *Proceedings of the 10th International Conference on the Foundations of Digital Games (FDG '15)*.
- [25] Martin Jennings-Teats, Gillian Smith, and Noah Wardrip-Fruin. 2010. Polymorph: Dynamic difficulty adjustment through level generation. (06 2010). DOI: <http://dx.doi.org/10.1145/1814256.1814267>
- [26] Jesper Juul. 2009. Fear of failing? the many meanings of difficulty in video games. *The Video Game Theory Reader* 2 (01 2009), 237–252.
- [27] Jesper Juul. 2013. The art of failure: an essay on the pain of playing video games.
- [28] Kari Kallinen. 2004. The effects of background music on using a pocket computer in a cafeteria: Immersion, emotional responses, and social richness of medium. In *In Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '04)*. ACM, New York, NY, USA, 1227–1230. DOI: <http://dx.doi.org/10.1145/985921.986030>
- [29] Dominic Kao and D. Harrell. 2016. Exploring the Impact of Avatar Color on Game Experience in Educational Games. In *2016 CHI Conference Extended Abstracts*. 1896–1905. DOI: <http://dx.doi.org/10.1145/2851581.2892281>
- [30] Oleksandra Keehl and Edward Melcer. 2019. Radical Tunes: Exploring the Impact of Music on

- Memorization of Stroke Order in Logographic Writing Systems. In *In Proceedings of the 13th International Conference on the Foundations of Digital Games (FDG 2019)*. DOI: <http://dx.doi.org/10.1145/3337722.3337764>
- [31] Brendan Keogh. 2013. When Game Over Means Game Over: Using Permanent Death to Craft Living Stories in Minecraft. In *In Proceedings of The 9th Australasian Conference on Interactive Entertainment: Matters of Life and Death*. DOI: <http://dx.doi.org/10.1145/2513002.2513572>
- [32] Mallory Ketcheson, Zi Ye, and T.C. Nicholas Graham. 2015. Designing for exertion: how heart-rate power-ups increase physical activity in exergames. In *In Proceedings of the 2015 Annual Symposium on Computer-Human Interaction in Play (CHI PLAY '15)*. 79–89. DOI: <http://dx.doi.org/10.1145/2513002.2513572>
- [33] Jan Klabbers. 2003. The gaming landscape: A taxonomy for classifying games and simulations. In *Digital Games Research Conference 2003*.
- [34] Lisbeth Klastrop. 2006. Why Death Matters: Understanding Gameworld Experience. In *In Proceedings of the International Conference on Advances in Computer Entertainment Technology (ACE '06)*. 29. DOI: <http://dx.doi.org/10.1145/1178823.1178859>
- [35] Christoph Klimmt, Christopher Blake, Dorothee Hefner, Peter Vorderer, and Christian Roth. 2009. Player Performance, Satisfaction, and Video Game Enjoyment. In *Entertainment Computing - 8th International Conference (ICEC '09)*, Vol. 5709. 1–12. DOI: http://dx.doi.org/10.1007/978-3-642-04052-8_1
- [36] Max Kreminski, Ben Samuel, Edward Melcer, and Noah Wardrip-Fruin. 2019. Evaluating AI-Based Games Through Retellings. In *In Proceedings of the Fifteenth AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment (AIIDE-19)*.
- [37] Filip Lange-Nielsen. 2011. The Power-up Experience: A study of Power-ups in Games and their Effect on Player Experience. In *In Proceedings of the Digital Games Research Association (DiGRA) Conference (DiGRA '11)*.
- [38] Raul Lara-Cabrera and David Camacho. 2018. A taxonomy and state of the art revision on Affective Games. *Future Generation Computer Systems* (01 2018). DOI: <http://dx.doi.org/10.1016/j.future.2017.12.056>
- [39] Alon Lessel. 2013. Nintendo Hard, or Hardly Working? (2013). <https://venturebeat.com/community/2013/09/27/nintendo-hard-or-hardly-working/>.
- [40] Chris Lewis, Jim Whitehead, and Noah Wardrip-Fruin. 2009. What Went Wrong: A Taxonomy of Video Game Bugs. In *In Proceedings of the 5th International Conference on the Foundations of Digital Games (FDG '09)*. DOI: <http://dx.doi.org/10.1145/1822348.1822363>
- [41] Derek Lomas, Kishan Patel, Jodi L. Forlizzi, and Kenneth R. Koedinger. 2013. Optimizing challenge in an educational game using large-scale design experiments. In *In Proceedings of the 2013 SIGCHI Conference on Human Factors in Computing Systems (CHI '13)*. DOI: <http://dx.doi.org/10.1145/2470654.2470668>
- [42] Rafael P. De Lope and Nuria Medina-Medina. 2016. A comprehensive taxonomy for serious games. *Journal of Educational Computing Research* 55, 5 (2016), 629–672. DOI: <http://dx.doi.org/10.1177/0735633116681301>
- [43] Ville Mäkelä, Sumita Sharma, Jaakko Hakulinen, Tomi Heimonen, and Markku Turunen. 2017. Challenges in Public Display Deployments: A

- Taxonomy of External Factors. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems (CHI '17)*. 3426–3475. DOI: <http://dx.doi.org/10.1145/3025453.3025798>
- [44] Simon Mccallum and Costas Boletsis. 2013. *Games for Health*. Chapter A Taxonomy of Serious Games for Dementia, 219–232. DOI: http://dx.doi.org/10.1007/978-3-658-02897-8_17
- [45] Edward Melcer and Katherine Isbister. 2016a. Bridging the Physical Divide: A Design Framework for Embodied Learning Games and Simulations. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems (CHI '16)*. DOI: <http://dx.doi.org/10.1145/2851581.2892455>
- [46] Edward Melcer and Katherine Isbister. 2016b. Motion, Emotion, and Form: Exploring Affective Dimensions of Shape. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems*. ACM, 1430–1437.
- [47] Edward Melcer and Marjorie Ann Cuerdo. 2020. *Game User Experience and Player-Centered Design*. Springer, Chapter Death & Rebirth in Platformer Games.
- [48] Diana Melnic and Vlad Melnic. 2018. Saved games and respawn timers: The dilemma of representing death in video games. *University of Bucharest Review: Literary and Cultural Studies Series VII* (07 2018), 29–37.
- [49] Florian Mueller, Martin Gibbs, and Frank Vetere. 2008. Taxonomy of exertion games. In *Proceedings of the 20th Australasian Computer-Human Interaction Conference (OZCHI '08)*. 263–266. DOI: <http://dx.doi.org/10.1145/1517744.1517772>
- [50] Souvik Mukherjee. 2009. 'Remembering How You Died': Memory, Death and Temporality in Videogames. In *Proceedings of the Digital Games Research Association (DiGRA) Conference (DiGRA '09)*. 263–266.
- [51] Lennart Nacke and Mark Grimshaw-Aagaard. 2010. Player-game interaction through affective sound. *Games Computing and Creative Technologies: Book Chapters* (2010). DOI: <http://dx.doi.org/10.4018/978-1-61692-828-5.ch013>
- [52] Simon Niedenthal. 2009. What We Talk About When We Talk About Game Aesthetics. In *Breaking New Ground: Innovation in Games, Play, Practice and Theory - Proceedings of DiGRA 2009 (DiGRA '09)*.
- [53] Jan Plass, Steffi Heidig, Elizabeth Hayward, Bruce Homer, and Enjoon Um. 2014. Emotional design in multimedia learning: Effects of shape and color on affect and learning. *Learning and Instruction* 29 (02 2014), 128–140. DOI: <http://dx.doi.org/10.1016/j.learninstruc.2013.02.006>
- [54] P.A. Rego, Pedro Moreira, and Luís Reis. 2010. Serious Games for Rehabilitation: A survey and a classification towards a taxonomy. In *5th Iberian Conference on Information Systems and Technologies (CISTI)*. 1–6.
- [55] Doris C. Rusch. 2009. Mechanisms of the Soul: Tackling the Human Condition in Videogames. In *Proceedings of the 2009 DiGRA International Conference: Breaking New Ground: Innovation in Games, Play, Practice and Theory (DiGRA '09)*. <http://hdl.handle.net/1721.1/100392>
- [56] Richard Ryan, C. Rigby, and Andrew Przybylski. 2006. The Motivational Pull of Video Games: A Self-Determination Theory Approach. *Motivation and Emotion* 30 (12 2006), 344–360. DOI: <http://dx.doi.org/10.1007/s11031-006-9051-8>
- [57] John Salisbury and Tom Cole. 2016. Grounded Theory in Games Research: Making the Case and

- Exploring the Options. In *International Joint Conference of DiGRA and FDG (DiGRA-FDG '16)*.
- [58] Timothy Sanders and Paul Cairns. 2010. Time perception, immersion and music in videogames. In *Proceedings of the 24th BCS Interaction Specialist Group Conference*. 160–167.
- [59] Anurag Sarkar and Seth Cooper. 2019. Transforming Game Difficulty Curves using Function Composition. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems (CHI '19)*. 160–167. DOI: <http://dx.doi.org/10.1145/3290605.3300781>
- [60] E.F. Schneider. 2004. Death with a Story: How Story Impacts Emotional, Motivational, and Physiological Responses to First-Person Shooter Video Games. 30 (2004), 361–375. Issue 3. DOI: <http://dx.doi.org/10.1093/hcr/30.3.361>
- [61] Kamran Sedig, Paul Parsons, and Robert Haworth. 2017. Player-Game Interaction and Cognitive Gameplay: A Taxonomic Framework for the Core Mechanic of Videogames. *Informatics* 4 (2017), 4. Issue 1. DOI: <http://dx.doi.org/10.3390/informatics4010004>
- [62] Noor Shaker, Julian Togelius, and Mark J Nelson. 2016. *Procedural content generation in games*. Springer.
- [63] Jan Smeddinck, Regan Mandryk, Max Birk, Kathrin Gerling, Dietrich Barsilowski, and Rainer Malaka. 2016. How to Present Game Difficulty Choices?: Exploring the Impact on Player Experience. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems (CHI '16)*. 5595–5607. DOI: <http://dx.doi.org/10.1145/2858036.2858574>
- [64] Adam Smith, Chris Lewis, Kenneth Hullett, Gillian Smith, and Anne Sullivan. 2011. An Inclusive Taxonomy of Player Modeling. (03 2011).
- [65] Gillian Smith and Jim Whitehead. 2007. A framework for analysis of 2D platformer levels. In *In Proceedings of the 2008 ACM SIGGRAPH Symposium on Video Games*. 75–80. DOI: <http://dx.doi.org/10.1145/1401843.1401858>
- [66] Nathan Sorenson, Philippe Pasquier, and Steve Dipaola. 2011. A Generic Approach to Challenge Modeling for the Procedural Creation of Video Game Levels. *IEEE Trans. Comput. Intellig. and AI in Games* 3 (09 2011), 229–244. DOI: <http://dx.doi.org/10.1109/TCIAIG.2011.2161310>
- [67] Zachary O. Toups, Jessica Hammer, William A. Hamilton, Ahmad Jarrah, William Graves, and Oliver Garretson. 2014. A framework for cooperative communication game mechanics from grounded theory. In *Proceedings of the first ACM SIGCHI annual symposium on Computer-human interaction in play* (pp. 257–266). ACM.
- [68] Eunjoon Um, Jan Plass, Elizabeth Hayward, and Bruce Homer. 2012. Emotional Design in Multimedia Learning. *Journal of Educational Psychology* 104 (05 2012), 485–498. DOI: <http://dx.doi.org/10.1037/a0026609>
- [69] D.P. Vossen. 2004. The nature and classification of games. *Avante* 10 (01 2004), 53–68.
- [70] Rina Wehbe, Elisa Mekler, Mike Schaeckermann, Edward Lank, and Lennart Nacke. 2017. Testing Incremental Difficulty Design in Platformer Games. In *In Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*. 5109–5113. DOI: <http://dx.doi.org/10.1145/3025453.3025697>
- [71] Mohammad Zohaib. 2018. Dynamic Difficulty Adjustment (DDA) in Computer Games: A Review. *Advances in Human-Computer Interaction* 4 (11 2018), 1–12. DOI: <http://dx.doi.org/10.1155/2018/5681652>

Ludography

- [G1] Adventures of Hendri. (Mar 7, 2018). *Developed by LionAnt.*
- [G2] Another Sight: Hodge's Journey. (Nov 14, 2018). *Developed by Lunar Great Wall Studios.*
- [G3] Ascendance. (Mar 27, 2018). *Developed by ONEVISION GAMES.*
- [G4] Away from Earth: Mars. (Aug 23, 2018). *Developed by Only Voxel Games.*
- [G5] Block Shock. (Feb 6, 2018). *Developed by VoxStudios.*
- [G6] Bloodstained: Curse of the Moon. (May 24, 2018). *Developed by INTI CREATES CO., LTD.*
- [G7] Bloody Trapland 2: Curiosity. (Feb 1, 2019). *Developed by 2Play Studios & Prasius.*
- [G8] Bombix. (Mar 2, 2018). *Developed by Pragmatix Ltd.*
- [G9] Bouncers. (Jun 1, 2018). *Developed by Firehawk Studios.*
- [G10] Castlevania Anniversary Collection. (May 16, 2019). *Developed by Konami Digital Entertainment.*
- [G11] Celeste. (Jan 25, 2018). *Developed by Matt Makes Games Inc.*
- [G12] Chamber of Darkness. (Oct10,2018). *Developed by The Crow Studios.*
- [G13] Chasm. (Jul 31, 2018). *Developed by Bit Kid, Inc.*
- [G14] Crash Bandicoot N. Sane Trilogy. (Jun 29, 2018). *Developed by Vicarious Visions & Iron Galaxy.*
- [G15] Cube - The Jumper. (May 15, 2018). *Developed by DZEJK.*
- [G16] Cube XL. (Mar 12, 2018). *Developed by Timberwolf Studios.*
- [G17] Cybarian: The Time Travelling Warrior. (Nov 9, 2018). *Developed by Ritual Games.*
- [G18] Dead Cells. (Aug 6, 2018). *Developed by Motion Twin.*
- [G19] Death's Gambit. (Aug 13, 2018). *Developed by White Rabbit.*
- [G20] DeepWeb. (Sep 20, 2018). *Developed by ImageCode.*
- [G21] Dream Alone. (Jun 28, 2018). *Developed by WarSaw Games.*
- [G22] Dungereed. (Feb 14, 2018). *Developed by TEAM HORAY.*
- [G23] Everything Will Flow. (Jul 28, 2018). *Developed by Hont.*
- [G24] Freezeer. (Jun 21, 2018). *Developed by NedoStudio.*
- [G25] Frog Demon. (Dec 11, 2018). *Developed by White Dog Games.*
- [G26] Golfing Over It with Alva Majo. (Mar 28, 201). *Developed by Majorariatto.*
- [G27] Gris. (Dec 13, 2018). *Developed by Nomada Studio.*
- [G28] Guacamelee! 2. (Aug 21, 2018). *Developed by DrinkBox Studios.*
- [G29] I was rebuilt. (Jun 28, 2018). *Developed by Gurila Ware Games.*
- [G30] Iconoclasts. (Jan 23, 2018). *Developed by Joakim Sandberg.*
- [G31] Katana ZERO. (Apr 18, 2019). *Developed by Askiisoft.*
- [G32] Lightform. (Feb 19, 2018). *Developed by Shadow Motion.*
- [G33] Little Marisa's Disaster Journey. (Apr 28, 2018). *Developed by Dark Sky Empire.*

- [G34] MagiCats Builder (Crazy Dreamz). (Jul 10, 2018). *Developed by Dreamz Studio.*
- [G35] Mark of the Ninja: Remastered. (Oct 9, 2018). *Developed by Klei Entertainment.*
- [G36] Mega Man 11. (Oct 2, 2018). *Developed by CAPCOM CO., LTD.*
- [G37] Mind Twins - The Twisted Co-op Platformer. (Jan 19, 2018). *Developed by DRUNKEN APES.*
- [G38] Mines of Mars. (Sep 10, 2018). *Developed by Wickey Ware.*
- [G39] Neon Beats. (May 3, 2019). *Developed by OKYO GAMES.*
- [G40] Niffelheim. (Sep 26, 2018). *Developed by Ellada Games.*
- [G41] Night Fly. (Jan 24, 2018). *Developed by ARGames.*
- [G42] Order No. 227: Not one step back!. (Jul 3, 2018). *Developed by High Wide.*
- [G43] Pogostuck: Rage with Your Friends. (Feb 28, 2019). *Developed by Hendrik Felix Pohl.*
- [G44] Poultry Panic. (Jan 17, 2018). *Developed by Virtual Top.*
- [G45] Razed. (Sep 14, 2018). *Developed by Warpfish Games.*
- [G46] ReCore: Definitive Edition. (Sep 14, 2018). *Developed by Armature Studio & Comcept.*
- [G47] Return. (Aug 3, 2018). *Developed by Breadmeat.*
- [G48] Richy's Nightmares. (Jul 10, 2018). *Developed by Unreal Gaming.*
- [G49] Rift Keeper. (Jan 14, 2019). *Developed by Frymore.*
- [G50] Riverhill Trials. (Apr 12, 2018). *Developed by Watercolor Games.*
- [G51] Running Man 3D. (Aug 21, 2018). *Developed by GGaming.*
- [G52] Slap City. (Mar 5, 2018). *Developed by Ludosity.*
- [G53] Steel Rats. (Nov 7, 2018). *Developed by Tate Multimedia.*
- [G54] Sure Footing. (Mar 30, 2018). *Developed by Table Flip Games.*
- [G55] The Cursed Tower. (Feb 6, 2018). *Developed by Mohsin Rizvi.*
- [G56] The Messenger. (Aug 30, 2018). *Developed by Sabotage.*
- [G57] Touhou Luna Nights. (Feb 25, 2019). *Developed by Vaka Game Magazine & Team Ladybug.*
- [G58] Trials of the Gauntlet. (Mar 16, 2018). *Developed by Broken Dinosaur Studios.*
- [G59] Trials Rising. (Feb 26, 2019). *Developed by RedLynx.*
- [G60] Vagante. (Feb 21, 2018). *Developed by Nuke Nine.*
- [G61] Viral Cry. (Mar 7, 2018). *Developed by Strategy Empire.*
- [G62] Wandersong. (Sep 27, 2018). *Developed by Greg Lobanov.*